

In this figure, which comes from simulated data, the distribution of the measurement values around the arithmetic mean is not symmetric, unlike a normal distribution. Consequently, the arithmetic mean is not at the center of the distribution on purpose and is shifted to the left. Indeed, it would be possible to compute the standard deviation, but it is not an appropriate estimate of MU. As explained before, the calculation principle of the standard uncertainty combines various standard deviations and uncertainties. The resulting statistical distribution of data is not necessarily Normal, as illustrated.

With the notations used so far, the coverage interval $[I_L; I_U]$ can be expressed in the following forms:

Diverse possible forms of the coverage interval

$$\begin{aligned} Z \pm k_{GUM} \times u(Z) \\ Z \pm U(Z) \\ Z \times (1 \pm UR\%) \end{aligned} \quad (7.32)$$

The uncertainty function can be used to calculate the coverage interval. It is satisfactory to replace the form of expression of MU with its function. At least three types of predictions can be made:

- The coverage interval of an unknown sample.
- The concentration that corresponds to a given MU.
- The concentration matches one of the bounds of coverage interval I_L or I_U .

7.6.2 Coverage Interval of Given Concentration

This first type of application is illustrated by the Resource N worksheet in the case where the uncertainty function is a power function. The application to a different kind of uncertainty function can easily be adapted. According to the convention, the calculation formulas appear in columns C and F. This is an application from THEOPHYLLINE data.

Resource N Coverage interval for a given concentration (Excel).						
	A	B	C	D	E	F
1	Resource N: Coverage interval for a given concentration					
2	Theophylline UR% function					
3	Constant	0.1813				
4	Power	-0.222				
5						
6	Concentration (µg/l)	UR%(X)		Coverage interval		
7	0.05	0.352554663	=B\$3*POWER(A7;\$B\$4)	0.03237227	0.06762773	=A7*(1+B7)
8	0.25	0.246635236	=B\$3*POWER(A8;\$B\$4)	0.18834119	0.31165881	=A8*(1+B8)
9	1.5	0.165693528	=B\$3*POWER(A9;\$B\$4)	1.25145971	1.74854029	=A9*(1+B9)
10	3	0.14206166	=B\$3*POWER(A10;\$B\$4)	2.57381502	3.42618498	=A10*(1+B10)
11	8	0.114264632	=B\$3*POWER(A11;\$B\$4)	7.08588294	8.91411706	=A11*(1+B11)

7.6.3 Coverage Interval of Given Relative Uncertainty

This second application is illustrated by the Resource O worksheet. It also applies to an uncertainty power function because it raises some programming issues with

Excel. To predict the concentration corresponding to a given relative uncertainty, it is necessary to go through the inverse of the power function, i.e. obtain the n th root of a number. The Excel built-in function `POWER` takes `number` and `power` for input arguments and returns the power raising of `number`. For example, the power argument 0.5, it is possible to replace the function `SQRT`, which only returns the square root of a number. But the inverse function is not available in Excel. The simplest way to solve this problem is to use logarithms (decimal in this case), as shown in the formulation of the inverse function below.

Power function

$$UR\% = c \times Z^d$$

Inverse of the power function

$$Z = 10^{\frac{\text{Log}(UR\%) - \text{Log}(c)}{d}} \quad (7.33)$$

The following Resource O worksheet illustrates this calculation and how to find the concentrations and their coverage intervals for various values of relative uncertainty. This type of prediction will be used in the Section 9.2 about a possible definition of the limit of quantification (LOQ). By comparing these results with Resource N worksheet, some differences are observable:

- Inputting 0.050 µg/l in row 7 of Resource N, the relative uncertainty of 35%.
- Inputting 35%, in line 7 of Resource O, the predicted concentration is 0.0517 µg/l.

These differences are due to the rounding of the coefficients of the model and results. If all the significant figures are kept, the results are identical.

Resource O Coverage interval for given relative uncertainty (Excel).						
	A	B	C	D	E	F
1	Resource O: Coverage interval for given relative uncertainty					
2	Theophylline UR% function					
3	Constant	0.181				
4	Power	-0.222				
5						
6	UR%(X)	Predicted concentration (µg/l)		Coverage interval		
7	35%	0.0517	=10^((LOG(A7)-LOG(\$B\$3))/(\$B\$4))	0.034	0.070	=\$B7*(1+\$A7)
8	30%	0.10	=10^((LOG(A8)-LOG(\$B\$3))/(\$B\$4))	0.072	0.134	=\$B8*(1+\$A8)
9	25%	0.24	=10^((LOG(A9)-LOG(\$B\$3))/(\$B\$4))	0.18	0.29	=\$B9*(1+\$A9)
10	20%	1	=10^((LOG(A10)-LOG(\$B\$3))/(\$B\$4))	0.51	0.77	=\$B10*(1+\$A10)
11	10%	15	=10^((LOG(A11)-LOG(\$B\$3))/(\$B\$4))	13	16	=\$B11*(1+\$A11)
12	5%	331	=10^((LOG(A12)-LOG(\$B\$3))/(\$B\$4))	315	348	=\$B12*(1+\$A12)

7.6.4 Obtain the Limits of the Coverage Interval

In Section 8.2, this last application of the uncertainty function is meaningful when dealing with sample conformity assessment. It consists in predicting the measurement value matching one of the two bounds of a pre-defined coverage interval. For instance, the official threshold contents of a pollutant in a food product must be

below a given value. It is necessary to know the MU at this threshold to verify if a control method is adequate. Two situations must be tackled, whereas the predicted measurement can be:

- The lower bound I_L of the coverage interval.
- The upper bound I_U of the coverage interval.

The two formulas below deal with these two cases.

Measurement at lower bound

$$Z_{min} = \frac{I_L}{1 - UR\%} \quad (7.34)$$

Measurement at upper bound

$$Z_{max} = \frac{I_U}{1 + UR\%} \quad (7.35)$$

The relative uncertainty $UR\%$ (Z) can be computed with the uncertainty function, and the application of the formulas is direct, as the following calculations show:

UR % (%)	Threshold (g/kg)		Acceptable values must be
40	Maximum limit	$I_U = 5.0$	≤ 3.57
10			≤ 4.55
40	Minimum limit	$I_L = 5.0$	≥ 8.33
10			≥ 5.56

Let us assume:

- The threshold value is 5 g/kg, representing either a maximum or a minimum bound.
- Two methods of analysis, one leading to 10% relative uncertainty for this content and the other 40%. The same values are used in the next chapter on sample conformity assessment and illustrated in Figure 8.4.

When using the previous formulas, it is now possible to define acceptable values when controlling a given threshold. For example, if the relative uncertainty is constant and equal to 40%, the highest acceptable concentration is 3.57 g/kg to satisfy the 5.0 g/kg threshold. This simple example with constant uncertainty is to illustrate the procedure, but the use of a power uncertainty function is not problematic.

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